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A numerical experiment on the stability and convergence characteristics of some splitting mixed-finite element methods for solving the incompressible Navier-Stokes equations

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Abstract In the present study, a fractional three-step P2P1 finite element method (FEM) for solving the unsteady incompressible Navier-Stokes equations, which is a variation of P1P1 four-step splitting FEM [1], was compared with conventional one-step time-integration schemes in terms of the CPU time and convergence characteristics of an iterative solver by the solution of some benchmark problems. One-step time-integration schemes were temporarily discretized by either the Crank-Nicolson or the Adams-Bashforth method. Fractional three-step P2P1 FEM consists of three steps: a non-linear momentum equation with the pressure in the previous time step is solved to obtain an intermediate velocity field by the Crank-Nicolson method in the first step and another intermediate velocity field is calculated using the pressure in the previous time step in the second step, and a divergence-free constraint is imposed in the last step to update the pressure field, in which a symmetric saddle-point type matrix (SPTM) is solved. It was shown that the fractional three-step method is more efficient than one-step time-integration schemes because a symmetric SPTM is assembled only once during the entire computation and solved once at each time-step; further, the cost of solving the nonlinear momentum equation in a fully-implicit manner is relatively low. Furthermore, the proposed method was found to be more stable than one-step time-integration schemes as it provided stable solutions at higher CFL numbers.

1. Introduction

Finite element method (FEM) is a valuable tool for solving the incompressible Navier-Stokes equations, and many investigators have shown that FEM can be successfully applied to various fluid flow problems. The major advantages of finite element formulations are that arbitrarily shaped domains are easily handled with unstructured meshes, and various adaption techniques based on h-refinement or p-refinement are provided to enhance the accuracy of the solutions of the incompressible Navier-Stokes equations. However, FEM has not seemed to be as popular as finite difference/volume methods in the field of computational fluid dynamics (CFD) presumably because FEM has been known to be expensive due to the requirement of a large memory and computational time. In a conventional one-step integrated FEM, the governing equations (momentum and continuity equations) are solved in a fully coupled way, and the velocity and pressure variables are simultaneously obtained.

To overcome this drawback, various splitting formulations, which are conceptually similar to those based on finite difference/volume methods, have been proposed in the context of *P1P1*

FEM. Since the velocity and pressure fields are obtained at each separate step in fractional step methods [1-6], a small and simple matrix is produced at each step compared to the integrated formulation. Also, the memory and cost requirements of the computations are significantly reduced because the pressure equation for mass conservation in the Navier-Stokes equations is constructed only once if the mesh is fixed. However, the splitting formulation solves the pressure Poisson equation of an elliptic type so that a pressure boundary condition is always needed along a boundary even when no physical information is given on the pressure. In general, open boundary problems sometimes have no information on either the pressure or the pressure gradient along the exit; further, both the velocity and the velocity gradient are unknown there. Therefore, the proper imposition of the outflow boundary condition is not clear, and this was an important consideration in the segregation approach [1-3]. Ramaswamy and Jue [2] suggested an additional pressure equation for the outflow boundary condition, and Choi et al. [1] suggested a pressure boundary condition at both the open and no-slip boundaries. Gresho [3] compared various boundary conditions for various splitting methods.

On the other hand, one-step time-integrated FEM has an advantage over the splitting formulation in the sense that it can yield more robust convergence behaviour because all the variables are implicitly coupled especially when the flow fields are complicated, as in strong swirl and impinging jet flows [7]. Also, one-step integrated FEM needs a small number of iterations to achieve convergence of an iterative solver compared to the splitting method for complicated flows. However, the assembly of a global matrix resulting from a one-step integrated FEM and the application of an ILU type preconditioner to the global matrix is not so straightforward as that of a small and simple matrix resulting from the splitting formulation; a saddle-point-type matrix (SPTM) [8] is generated when a *P2P1* mixed finite element that satisfies the Brezzi condition [9] is used. As reported in the Refs. [10, 11], the FEM based on *P2P1* formulation requires to solve a coupled system of equations with velocity and pressure variables that leads to the saddle point type matrix to satisfy the divergence free constraint in a rigorous manner. The SPTM is assembled and preconditioned at every time-step when the momentum equation is solved implicitly; therefore, it needs a large computing time.

As mentioned above, *P1P1* splitting and one-step integrated FEM have not only advantages but also disadvantages. To make up for the weak-points in *P1P1* splitting and one-step integrated FEM, various multi-step mixed finite element methods have been proposed. These multi-step mixed FEMs can utilize the fact that an SPTM for the continuity constraint is assembled and preconditioned only once as in the *P1P1* splitting method [1-6] during the entire computation as long as the mesh does not change because the momentum (convection and diffusion) equation is split from the continuity equation. A three-step operator splitting *P2P1* (quadratic velocity/linear pressure) FEM was proposed by Choi and successfully employed for the direct numerical simulation of a fluid-particle

mixture [12]. Although various multi-step mixed FEMs have been proposed for the unsteady incompressible Navier-Stokes equations, a comparison of multi-step FEM with one-step FEM has not been reported yet in terms of the CPU time, convergence characteristics of an iterative solver, and stability. Therefore, we compare the fractional three-step *P2P1* FEM and one-step integration FEM for well-known benchmark problems: one-sided and two-sided lid-driven cavities, a decaying vortex, and an unsteady flows past a circular cylinder at $Re = 100$.

2. Numerical method

In this section, the governing equations of the unsteady incompressible Navier-Stokes equations are described with the boundary conditions used in this study. Then, two one-step finite element formulations and a fractional three-step formulation are described in detail.

2.1 Governing equations

The governing equations are the continuity and unsteady incompressible Navier-Stokes equations, which are written using indicial notation as follows:

$$\begin{aligned} u_{i,j} &= 0, \\ \rho(\dot{u}_i + u_j u_{i,j}) &= -p_{,i} + (\tau_{ij})_{,j}, \end{aligned} \quad (1)$$

where ‘ $\dot{\cdot}$ ’ denotes the time derivative, ‘ $,$ ’ the spatial derivatives, u_i the i^{th} component of the velocity, p the pressure, ρ the density, and τ_{ij} the shear stress, which is expressed as follows:

$$\tau_{ij} = \mu(u_{i,j} + u_{j,i}). \quad (2)$$

In Eq. (2), μ denotes the dynamic viscosity. The boundary conditions for Eq. (1) are written as follows:

$$u_i|_{\Gamma_1} = b_i, \quad \sigma_{ij} n_j|_{\Gamma_2} = t_i, \quad (3)$$

where Γ_1 and Γ_2 are two non-overlapping subsets of the piecewise-smooth domain boundary Γ , b_i is the velocity vector prescribed on Γ_1 , $\sigma_{ij} (= -p\delta_{ij} + \tau_{ij})$ is the stress tensor, t_i is the traction vector prescribed on Γ_2 , and n_j is the outward unit vector normal to Γ_2 . In the present study, a *P2P1* finite element that satisfies the Brezzi condition [9] was employed for all the formulations.

Fig. 1 shows the *P1P1* and *P2P1* elements. The *P1P1* element allocates both the velocity and the pressure at vertex nodes, whereas the *P2P1* element allocates the pressure at vertex nodes and the velocity at both vertex and mid nodes. Therefore, in a three-dimensional problem, compared with the mesh based on *P1P1* elements, the number of pressure variables in the mesh based on *P2P1* elements is reduced to approximately an eighth of that for *P1P1* elements.

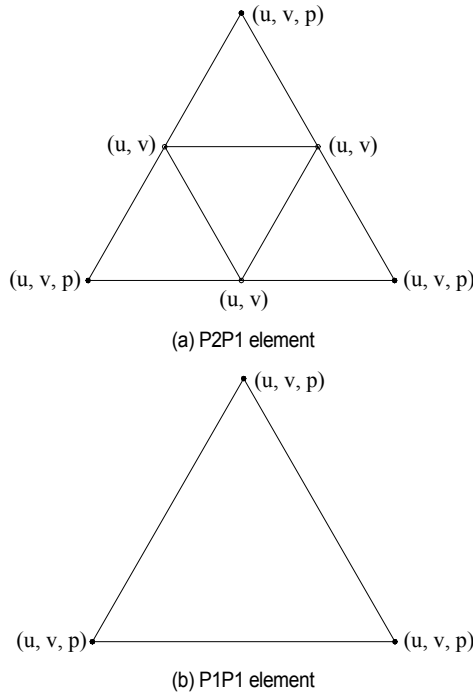


Fig. 1. The velocity-pressure elements used.

2.2 One-step crank-nicolson method (method-I)

The governing equations of one-step integrated FEM based on the Crank-Nicolson method are written as follows:

$$u_{i,j}^{n+1} = 0, \quad (4)$$

$$\rho \frac{u_i^{n+1} - u_i^n}{\Delta t} + \rho \frac{1}{2} (u_j^{n+1} u_{i,j}^{n+1} + u_j^n u_{i,j}^n) = -p_i^{n+1} + \frac{1}{2} (\tau_{ij}^{n+1} + \tau_{ij}^n),$$

where Δt is the time increment and the superscript n denotes the time level. The convection and diffusion terms are integrated using the Crank-Nicolson method, and the weak formulation of Eq. (4) is written as follows:

Find $u_i^{n+1} \in H^1(\Omega)$ and $p^{n+1} \in L^2(\Omega)$ such that

$$\int_{\Omega} \left\{ W_i \rho \left[\frac{u_i^{n+1} - u_i^n}{\Delta t} + \frac{1}{2} (u_j^{n+1} u_{i,j}^{n+1} + u_j^n u_{i,j}^n) \right] + W_{i,j} (-p^{n+1} \delta_{ij} + \tau_{ij}^{n+1}) \right\} d\Omega$$

$$- \int_{\Gamma_2} W_i (-p^{n+1} \delta_{ij} + \tau_{ij}^{n+1}) n_j d\Gamma_2 + \int_{\Omega} q u_{i,j}^{n+1} d\Omega = 0, \quad (5)$$

for all $W_i \in V$ and $q \in P$, $V = \{W_i | W_i \in H^1(\Omega), W_i = 0 \text{ on } \Gamma_1\}$, $P = \{q | q \in L^2(\Omega)\}$.

In the above, W_i is the weighting function vector for the momentum equation, Γ is the boundary of Ω , n_j is the outward unit normal vector to Γ , and q is the weighting function for the continuity equation. The Galerkin method is employed for spatial discretization, and a fully-implicit Crank-Nicolson method is adopted for temporal discretization. Then, the following saddle-point type global matrix is obtained:

$$\begin{pmatrix} A & B \\ B^T & 0 \end{pmatrix} \begin{pmatrix} u \\ p \end{pmatrix} = \begin{pmatrix} f \\ 0 \end{pmatrix}. \quad (6)$$

The saddle-point type matrix given in Eq. (6) is solved by bi-CGSTAB (bi-conjugate gradient stabilized) [13] with an adapted ILU (AILU) preconditioner [8, 14]. The successive-substitution method [15] is used to treat the nonlinearity of the momentum equations.

2.3 One-step adams-bashforth method (method-II)

The governing equations of one-step integrated FEM based on the Adams-Bashforth method are written as follows:

$$u_{i,j}^{n+1} = 0, \quad (7)$$

$$\rho \frac{u_i^{n+1} - u_i^n}{\Delta t} + \rho \left(\frac{3}{2} u_j^n u_{i,j}^n - \frac{1}{2} u_j^{n-1} u_{i,j}^{n-1} \right) = -p_i^{n+1} + \tau_{ij}^{n+\frac{1}{2}}.$$

The weak formulation of Eq. (7) is derived as follows by multiplying it by a weighting function and integrating over the spatial domain as in the one-step Crank-Nicolson method:

Find $u_i^{n+1} \in H^1(\Omega)$, $p^{n+1} \in L^2(\Omega)$ such that

$$\int_{\Omega} \left\{ W_i \rho \left[\frac{u_i^{n+1} - u_i^n}{\Delta t} + \left(\frac{3}{2} u_j^n u_{i,j}^n - \frac{1}{2} u_j^{n-1} u_{i,j}^{n-1} \right) \right] + W_{i,j} (-p^{n+1} \delta_{ij} + \tau_{ij}^{n+\frac{1}{2}}) \right\} d\Omega$$

$$- \int_{\Gamma_2} W_i (-p^{n+1} \delta_{ij} + \tau_{ij}^{n+\frac{1}{2}}) n_j d\Gamma_2 + \int_{\Omega} q u_{i,j}^{n+1} d\Omega = 0, \quad (8)$$

for all $W_i \in V$ and $q \in P$, $V = \{W_i | W_i \in H^1(\Omega), W_i = 0 \text{ on } \Gamma_1\}$, $P = \{q | q \in L^2(\Omega)\}$.

The Galerkin method is employed for spatial discretization, and the Adams-Bashforth method is adopted for temporal discretization. Although the global matrix generated from Eq. (8) is a symmetric SPTM, it is solved by Bi-CGSTAB as in method-I for comparison.

2.4 A fractional three-step method (Method-III)

The governing equations of the fractional three-step integrated finite element method used in the present study are written as follows:

$$\rho \left[\frac{\hat{u}_i - u_i^n}{\Delta t} + \frac{1}{2} (\hat{u}_j \hat{u}_{i,j} + u_j^n u_{i,j}^n) \right] = -p_i^n + \frac{1}{2} (\hat{\tau}_{ij} + \tau_{ij}^n), \quad (9a)$$

$$\rho \frac{u_i^* - \hat{u}_i}{\Delta t} = p_i^n, \quad (9b)$$

$$\begin{cases} \rho \frac{u_i^{n+1} - u_i^*}{\Delta t} = -p_i^{n+1}, \\ u_{i,j}^{n+1} = 0. \end{cases} \quad (9c)$$

In the first and second fractional steps, the method solves

the same equations as those of the fractional four-step *P1P1* formulation [1] to satisfy the momentum equation. The successive-substitution method is used to treat the nonlinearity of the momentum equations. For time integration, the Crank-Nicolson scheme, which is second-order accurate and unconditionally stable, is used. The second step is necessary for the second-order accuracy of the boundary condition of the intermediate velocity obtained in Eq. (9a) [1, 16]. The weak formulation of the first step can be stated as follows:

Find $\hat{u}_i \in H^1(\Omega)$ such that

$$\int_{\Omega} \left\{ W_i p \left[\frac{\hat{u}_i - u_i^n}{\Delta t} + \frac{1}{2} (\hat{u}_i \hat{u}_{i,j} + u_i^n u_{i,j}^n) \right] + W_{i,j} \left[-p^n \delta_{ij} + \frac{1}{2} (\hat{\tau}_{ij} + \tau_{ij}^n) \right] \right\} d\Omega - \int_{\Gamma_2} W_i \left[-p^n \delta_{ij} + \frac{1}{2} (\hat{\tau}_{ij} + \tau_{ij}^n) \right] n_j d\Gamma_2 = 0, \quad (10)$$

for all $W_i \in V$, $V = \{W_i | W_i \in H^1(\Omega), W_i = 0 \text{ on } \Gamma_1\}$.

The asymmetric matrix generated from Eq. (10) is solved using bi-CGSTAB, and the successive-substitution method [15] is used to treat the nonlinearity of the momentum equations as in method-I. In order to utilize the fact that a separate pressure equation, which is assembled and preconditioned only once, is to be solved as in the *P1P1* fractional step method [1], a fixed symmetric SPTM is obtained for the third step as in method-II. The Galerkin formulation of Eq. (9c) is written as follows:

Find $u_i^{n+1} \in H^1(\Omega)$, $p^{n+1} \in L^2(\Omega)$ such that

$$\int_{\Omega} \left[W_i p \left(\frac{u_i^{n+1} - u_i^n}{\Delta t} \right) + W_{i,j} (-p^{n+1} \delta_{ij}) \right] d\Omega + \int_{\Gamma_2} W_i p^{n+1} \delta_{ij} n_j d\Gamma_2 + \int_{\Omega} q u_{i,i}^{n+1} d\Omega = 0, \quad (11)$$

for all $W_i \in V$ and $q \in P$, $V = \{W_i | W_i \in H^1(\Omega), W_i = 0 \text{ on } \Gamma_1\}$, $P = \{q | q \in L^2(\Omega)\}$.

Because we use the *P2P1* element in the present Galerkin formulation, W_i and q are second-order and first-order polynomials, respectively. The *P1P1* splitting method [1-6] solves the pressure Poisson equation, and then corrects the intermediate velocity using the pressure field obtained from the continuity constraint. Therefore, it has the demerit of requiring an extra pressure boundary condition along a boundary even when no physical information is given on the pressure. For some solutions to this problem, one can refer to Refs. [1-3, 5, 6]. On the other hand, only one reference pressure value needs to be given in the mixed *P2P1* finite element formulation since it solves for the velocities and pressure simultaneously. And as mentioned above, the SPTM in the third step needs to be assembled and preconditioned only once during the entire computation if the mesh is fixed. The SPTM generated from Eq. (11) is solved by bi-CGSTAB as in method-I for comparison with the other methods. One of the important advantages of the fractional-step method is that an optimal preconditioner among various ones can be adopted to efficiently solve each step of equation. Recently, Nguyen et al. [17] examined the perform-

ances of various parallel ILU(0) preconditioners based on the domain decomposition method for the *P2P1* formulation of the incompressible Navier-Stokes equations to solve coronary blood flow. Lee et al. [18] investigated the performance of a parallel ILU(0) preconditioner based on the domain decomposition and message passing interface (MPI) method [19] implemented for the CUPID [20] code.

2.5 Preconditioning method

An ILU-type preconditioner has been effectively used together with iterative solvers to solve the pressure Poisson equation in incompressible computational fluid dynamics problems. However, the zero-pivot problem may occur in the course of ILU-type operations for the SPTM unless proper care is taken since the SPTM obtained from *P2P1* finite element formulations generates a zero-block, as in Eq. (6) [8, 14].

Therefore, an AILU preconditioner [8] is employed for all the computations of the present study.

3. Results and discussions

All computations in this section were performed on a personal computer (Intel(R) core(TM) 2 Duo CPU E8400 3 GHz, memory of 6 GByte, Redhat Enterprise 2.6.9-55 Operating System, and Intel(R) Fortran 10.1 Compiler). In the first two subsections, a lid-driven cavity flow is tested for validating the methods used in this study, and a decaying vortex problem is solved to check the spatial accuracy of the three methods. In the subsequent sections, unsteady flows behind a two-dimensional circular cylinder and a sphere are investigated to compare the three methods in terms of their performance.

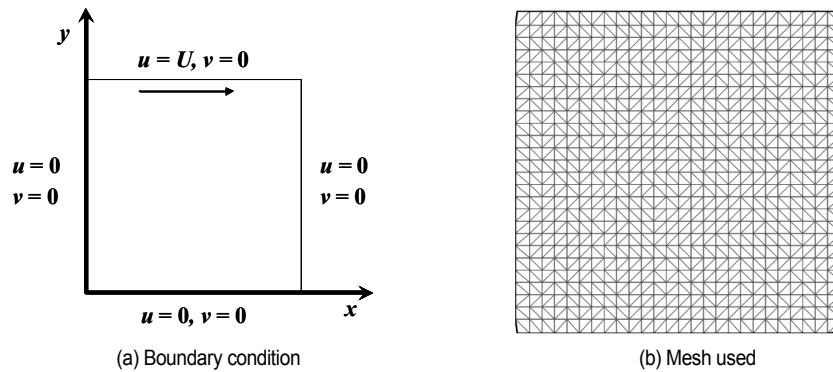
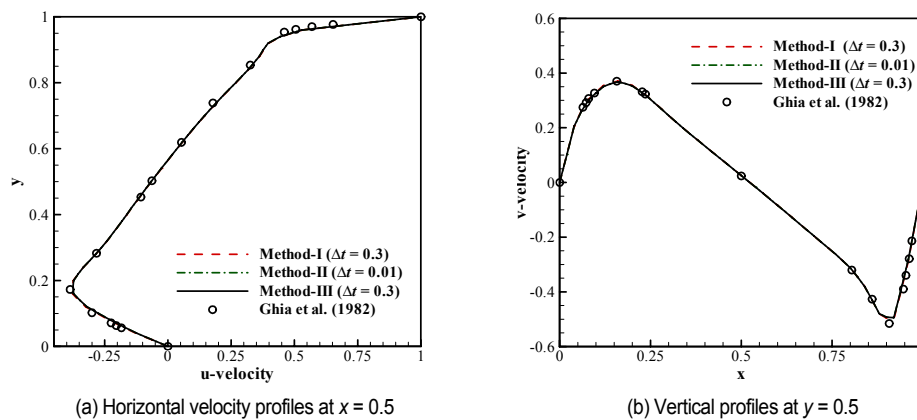
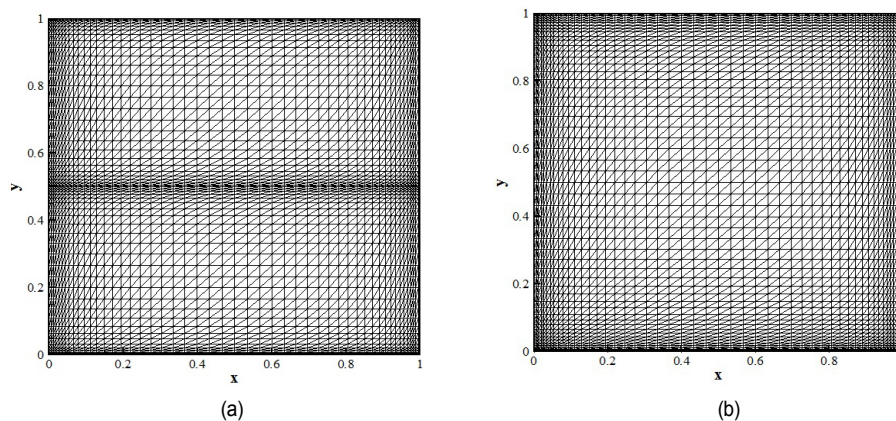
For the iterative solver used in this study, a residual vector is defined as follows:

$$r = b - [A]x. \quad (12)$$

Every simulation is terminated when the relative L_2 norm of the residual vector $\|r\| / \|r_0\|$ is less than 10^{-6} .

3.1 Lid-driven cavity

The lid-driven cavity problem at $Re = 1000$ was tested as a benchmark problem in order to validate the three methods used in this study. The fluid contained inside a squared cavity is set into motion by the upper wall, which is sliding at constant speed. The domain and boundary conditions for the one-sided lid-driven cavity and the mesh used for the present simulation are shown in Fig. 2. The overall numbers of nodes, elements, and variables in the mesh are 2809, 1352, and 6347, respectively. The Reynolds number is defined as $Re = UL/\nu$, where U is the velocity of the top lid, $L = 1$ m is used as the length of the cavity side wall, and ν is the kinematic viscosity of the fluid. The velocity boundary conditions are $u = U = 1$ m/s and $v = 0$ on the

Fig. 2. Schematic description for the one-sided lid-driven cavity flow at $Re = 1000$.Fig. 3. Comparison of (a) horizontal velocity profiles at $x = 0.5$; (b) vertical velocity profiles at $y = 0.5$.Fig. 4. Finite element mesh used for the two-sided lid-driven cavity flow at $Re = 400$: (a) parallel wall motion; (b) anti-parallel wall motion.

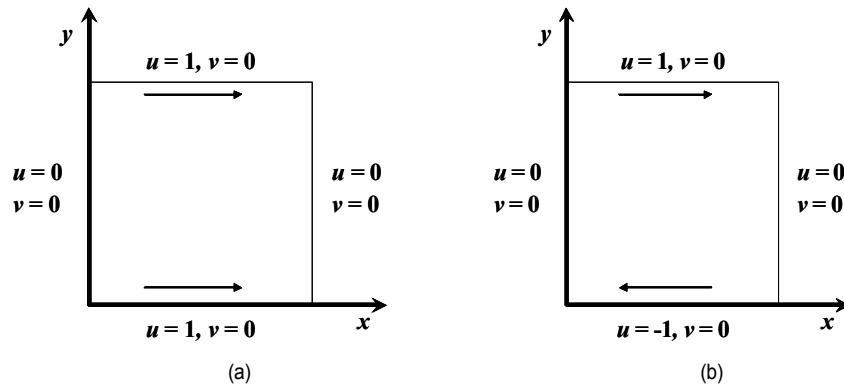
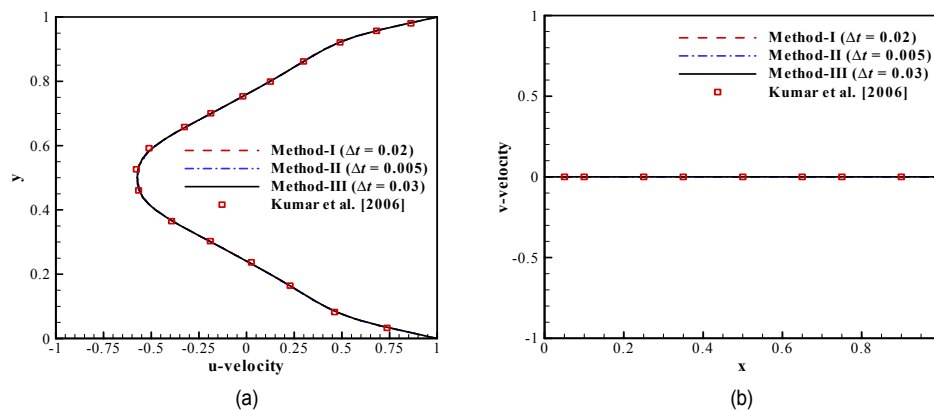
top lid, and $u = v = 0$ on the walls. Three time-step sizes ($\Delta t = 0.01, 0.1$, and 0.3 s) were used to investigate the temporal accuracy and stability of the three methods. The maximum allowable time-steps and CFL numbers are 0.3 and 17.22 for methods I and III and 0.01 and 0.573 for Method II, respectively. Both method-I and method-III are stable, generating accurate solutions under CFL numbers up to 17 , whereas method-II diverges under CFL numbers larger than 1.0 . Figs. 3(a) and (b) show the horizontal velocity profiles at $x = 0.5$ and vertical profiles at $y = 0.5$, respectively, as obtained by the

three methods. The three results agree well with that of Ghia et al. [21] under a relatively coarse mesh.

Next, two-sided lid-driven cavity problems were investigated at $Re = 400$. Figs. 4(a) and (b) denote the finite-element mesh used for parallel and anti-parallel wall motions, respectively. In the case of the two-sided lid-driven cavity, the overall numbers of nodes, elements, and variables in the mesh were 10201 , 5000 , and 23003 , respectively. The parallel wall motion for which the top and bottom walls are moving in the same direction and the anti-parallel wall motion for which the top and bot-

Table 1. CFL numbers for the two-sided lid-driven cavity problem with parallel wall motion.

Method	CFL number		
	$\Delta t = 0.005$	$\Delta t = 0.02$	$\Delta t = 0.03$
Method-I	1.84	7.35	X
Method-II	1.84	X	X
Method-III	1.84	7.35	11.03

Fig. 5. Schematic description for the two-sided lid-driven cavity flow at $Re = 400$: (a) parallel wall motion with boundary conditions; (b) anti-parallel wall motion with boundary conditions.Fig. 6. For parallel wall motion: (a) horizontal velocity profiles at $x = 0.5$; (b) vertical velocity profiles at $y = 0.5$.

tom walls are moving in the reverse direction are shown in Figs. 5(a) and (b), respectively.

As in the lid-driven cavity problem, three time-steps were used to investigate the temporal accuracy and stability of the three methods for the two-sided lid-driven cavity problems. Table 1 shows that method-III is more stable than the other methods, giving an accurate solution under a CFL number larger than 10 whereas the other methods diverge with the CFL number for the cavity problem with parallel wall motion. method-II diverges under time-steps larger than 0.005 because of the time-step limit imposed for the semi-implicit method. For the cavity problem with anti-parallel wall motion, Table 2 also shows that method-III is more stable than the other methods. Figs. 6 and 7 show the horizontal velocity profiles at $x = 0.5$ and vertical profiles at $y = 0.5$, respectively, for parallel and anti-parallel wall motions, respectively, at a steady state. As

shown in Figs. 6 and 7, the results are in good agreement with the results of Kumar et al. [22].

3.2 Decaying vortex

To validate the spatial accuracy of the proposed methods, the three methods were applied to the following two-dimensional decaying vortex problem:

$$u(x, y, t) = -\cos \pi x \cdot \sin \pi y \cdot e^{-2\pi^2 t / Re}, \quad (13)$$

$$v(x, y, t) = \sin \pi x \cdot \cos \pi y \cdot e^{-2\pi^2 t / Re}, \quad (14)$$

$$p(x, y, t) = -0.25(\cos 2\pi x + \cos 2\pi y)e^{-4\pi^2 t / Re}. \quad (15)$$

The computational domain is $-1/2 \leq (x, y) \leq 1/2$, and computations were carried out at $Re = 10$, where $Re = U_1 L / \nu$, U_1 is the

Table 2. CFL numbers for the two-sided lid-driven cavity problem with anti-parallel wall motion.

Method	CFL number		
	$\Delta t = 0.005$	$\Delta t = 0.01$	$\Delta t = 0.02$
Method-I	1.86	3.73	X
Method-II	1.86	X	X
Method-III	1.86	3.73	7.44

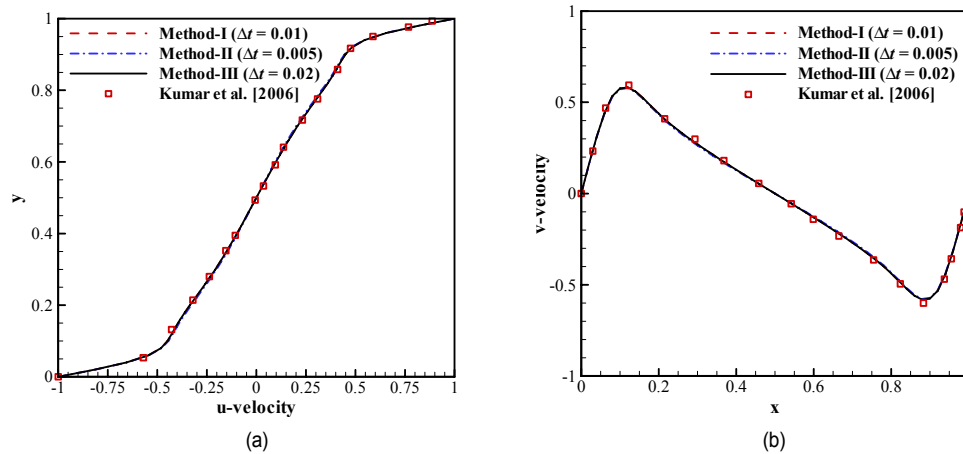
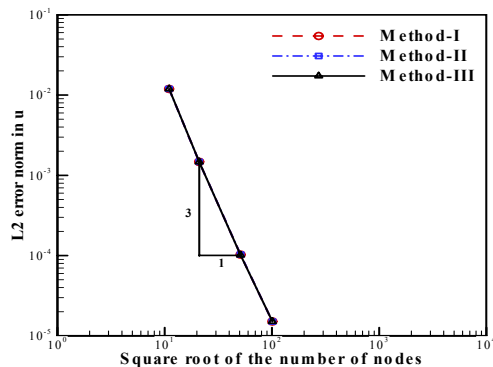


Fig. 7. For anti-parallel wall motion: (a) horizontal velocity; (b) vertical velocity.

Fig. 8. L2 error norm in u at $t = 0.3$ s for the three methods with $P2P1$ elements.

initial maximum velocity, and L is the size of the vortex. The initial velocity condition at $t = 0.0$ and the velocities at the boundaries over time are provided from the exact solution. Four different uniform meshes with triangular elements were tested to determine the spatial accuracy.

The numbers of nodes used in the calculations were 121 (11×11), 441 (21×21), 2601 (51×51), and 10201 (101×101), respectively. The L_2 error norm in u is defined as follows:

$$\|e\| = \sqrt{\sum_{i=1}^n (u(i) - u_{\text{exact}}(i))^2}. \quad (16)$$

Fig. 8 shows the variation of the L_2 error norm with the element size at $t = 0.3$ s when an identical time-step ($\Delta t = 0.001$)

is used. Note that the time-step used in this case is a small value ($\Delta t = 0.001$) such that the temporal error has a negligible effect on the total error [23]. Fig. 8 shows that all the three methods using the $P2P1$ element are almost third-order accurate [24].

3.3 Flow behind a circular cylinder at $Re = 100$

Next, we consider flow past a circular cylinder. In this problem, we have a uniform upstream flow, where the Reynolds number based on the cylinder diameter and the upstream velocity is 100. The flow domain, boundary conditions, and unstructured mesh used in the present simulation are shown in Fig. 9. The dimensions of the computational domain, normalized by the cylinder diameter, are 37 and 56 in the streamwise and cross-stream directions, respectively. The ICEM-CFD package was used for unstructured finite element mesh generation, and the finite element mesh consisted of 6846 triangular elements with 3490 vertical points and 13826 nodal points, respectively.

The CPU times and iteration numbers of the iterative solver are examined for the three methods in Table 3 at the first and hundredth time-steps. Three different time-steps were used to investigate the numerical stability of the three methods.

In Table 3, the integer Iter1 denotes the iteration number of the iterative Bi-CGSTAB solver for solving a non-symmetric SPTM in Method-I, a symmetric SPTM in method-II, and a non-symmetric matrix of the first step in method-III. The integer Iter2 denotes the non-linear iteration number of method-I and the

Table 3. Comparison of CPU times and iteration numbers to convergence of the iterative solver of the three methods at the first and the hundredth time step computations of the unsteady flow past a circular cylinder at $Re = 100$ when three different time-steps are used.

	1st time step			100th time step		
Δt	0.02 s	0.05 s	0.10 s	0.02 s	0.05 s	0.10 s
CPU time (Iter1, Iter2) of method-I	16.66 s (130, 8)	21.24 s (132, 10)	22.8 s (119, 12)	6.53 s (135, 3)	4.21 s (86, 3)	5.8 s (89, 4)
CPU time (Iter1) of method-II	2.05 s (201)	X	X	0.79 s (76)	X	X
CPU time (Iter1, Iter2, Iter3) of method-III	3.20 s (4, 7, 189)	2.39 s (4, 9, 132)	2.56 s (6, 13, 132)	1.50 s (4, 3, 87)	1.36 s (3, 4, 78)	1.35 s (6, 4, 71)
	2.96 s	2.09 s	2.08 s	1.37 s	1.23 s	1.13 s

*The first row of the method-III denotes the total elapsed time for the completion of one time step whereas the second row the elapsed time for solving the SPTM in the third step.

Table 4. Comparison of CPU times of the three methods for the simulation of the unsteady flow past a circular cylinder at $Re = 100$ from $tU/d = 0.0$ to $tU/d = 10.0$ when three different time step size are used.

Time step size (CFL no.)	0.02 s (0.7)	0.05 s (1.74)	0.10 s (3.5)
Method-I	2582.2 s	1128.6 s	772.3 s
Method-II	446.4 s	X	X
Method-III	772.9 s	301.8 s	161.4 s

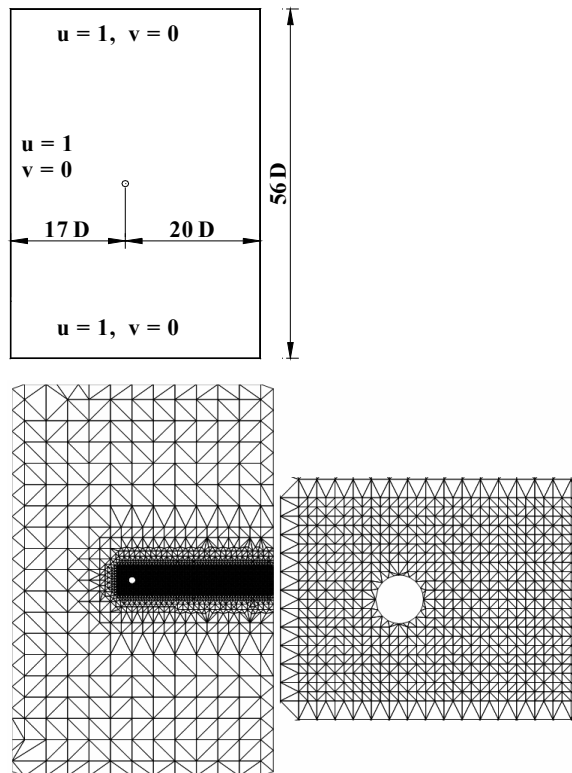


Fig. 9. Non-uniform grid system and the boundary conditions for the flow past a circular cylinder.

first step of method-III, and the integer Iter3 denotes the iteration number of the Bi-CGSTAB solver for solving the SPTM in the third step of method-III.

Table 3 shows that Iter1 of method-III is much smaller than those of the other methods, whereas Iter2 of method-III is

comparable to that of method-I regardless of the time step used. It also shows that Iter3 of method-III is comparable to Iter1 of method-I regardless of the time step and close to Iter1 of method-II for a small time-step. Table 3 also shows that the CPU time of method-III is mostly consumed for solving the SPTM in the third step, which is solved only once in a time-step. Therefore, method-III is faster than method-I for all time-steps and more stable than method-II, for which only a small time-step can be used due to the CFL limit.

Table 4 compares the CPU times used in the three methods for the simulation from $tU/d = 0.0$ to $tU/d = 10.0$ for three time-steps. Overall, it can be said that method-III is more efficient than the other methods in terms of the CPU time and stability.

As the Reynolds number is increased beyond 40, the flow behind a circular cylinder becomes unstable and vortex shedding occurs. The evolution was measured at some point of wake region behind the cylinder. Fig. 10 shows that periodic flow patterns appear behind the cylinder, and the vortex shedding periods are nearly the same as $6.14D/u_\infty$ for the three methods. The Strouhal numbers obtained from the three methods are compared with previous numerical [25-28] results and the experimental result of Williamson [25] in Table 5. The three methods used in the present study give nearly the same Strouhal numbers and agree well with the experimental result [25]. The time histories of the maximum allowable CFL numbers are shown in Fig. 11 for the three methods. $\Delta t = 0.2$ (CFL No.~7) could be used for method-I and method-III, whereas $\Delta t = 0.02$ (CFL No.~0.7) could be used for method-II.

3.4 Flow past a sphere at $Re = 300$

Finally, we considered the flow of a viscous fluid past a sphere at $Re = 300$ (the Reynolds number is based on the inlet

Table 5. Comparison of the present results with those of others for the flow behind a cylinder ($Re = 100$).

References	Method	Strouhal number, St
Present study	Method-I	0.163
	Method-II	0.163
	Method-III	0.163
Williamson [25]		0.164
Braza et al. [26]	FVM (13530 grid points)	0.160
Karniadakis et al. [27]	Spectral element method	0.179
Kwon and Choi [28]	FDM (321×321)	0.165

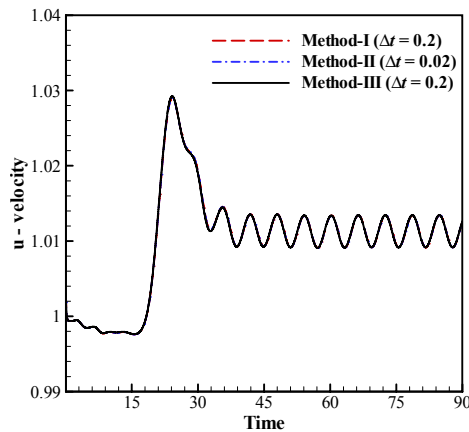


Fig. 10. Temporal evolution of the velocity at a point behind a circular cylinder.

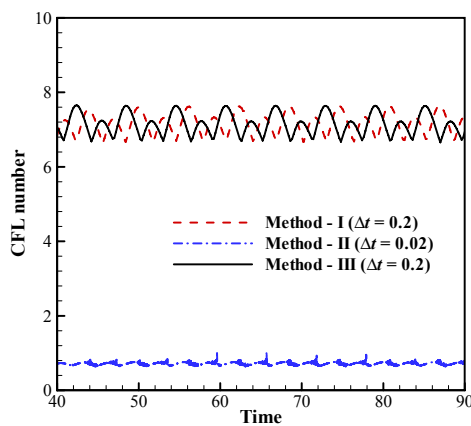


Fig. 11. Temporal evolution of the maximum CFL number for the three methods.

fluid velocity and diameter of the sphere) using the three methods. Unlike two-dimensional flows, three-dimensional flows are capable of admitting even more complicated kinematic and vortical interactions. The onset of unsteadiness is known to occur at a Reynolds number in the range of 270 to 300 [29]. A cross-sectional view of an unstructured mesh on the symmetric plane is shown in Fig. 12. The ICEM-CFD package was used for unstructured mesh generation. The overall numbers of nodes, vertex nodes, and elements were 151344, 19438, and 111532, respectively. Four time-steps ($\Delta t = 0.005, 0.01, 0.025,$

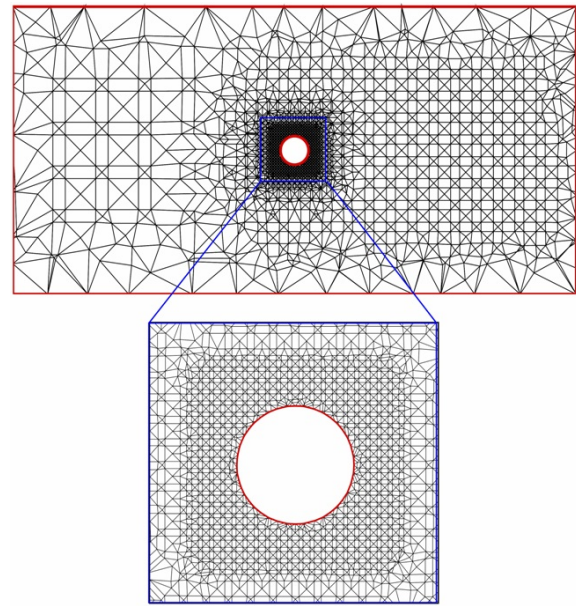


Fig. 12. 3-dimensional unstructured mesh around a sphere on the x-y plane.

and 0.05) were used. Fig. 13 shows the instantaneous streamlines at $Re = 300$. Fig. 13(a) shows that the (x, y)-plane retains a symmetric pattern. In contrast, Fig. 13(b) shows an asymmetric vortex shedding on the (z, x)-plane.

Fig. 14 shows the time histories of the pressure at a point behind a sphere at $Re = 300$ for the three methods. It is clear from Fig. 14 that a periodic flow pattern appears behind the sphere. The corresponding Strouhal number was found to be 0.138 for every scheme and time-step. This also agrees well with the value of 0.137 found by the numerical experiments of Johnson and Patel [29].

Lastly, the CPU times and iteration numbers of the iterative solver are also examined as in the case of the flow past a circular cylinder for the three methods in Table 6. The definitions of Iter1, Iter2, and Iter3 are the same as those in Table 3. As shown in Table 6, the trends are almost similar to those of Table 3 except that the CPU time used in the first step of method-III is comparable to that in the third step and that method-I is not as slow as in the two-dimensional case shown in Table 3. Table 7 compares the CPU times used in the three methods for the simulation of unsteady flow past a sphere at $Re = 300$

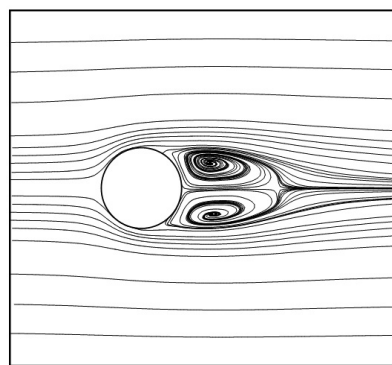
Table 6. Comparison of CPU times (in second) and iteration numbers to convergence of the iterative solver of the three methods at the first and the hundredth time step computations of the unsteady flow past a sphere at $Re = 300$ when four different time step size are used.

Δt	1st time step				100th time step			
	0.005	0.01	0.025	0.05	0.005	0.01	0.025	0.05
CPU time (Iter1, Iter2) of method-I	99.9 (64,5)	129.9 (61,6)	144.5 (58,8)	X	62.6 (57,3)	53.2 (56,3)	81.62 (51,4)	X
CPU time (Iter1) of method-II	17.7 (78)	X	X	X	11.8 (46)	X	X	X
CPU time (Iter1, Iter2, Iter3) of method-III	43.4 (11, 7, 80)	39.7 (10, 8, 66)	38.9 (9, 10, 62)	42.3 (9, 10, 59)	30.8 (10, 3, 48)	31.4 (10, 3, 49)	40.46 (10, 4, 52)	41.9 (10, 4, 56)
	26.3	21.8	17.5	20.1	16.2	16.5	20.9	22.5

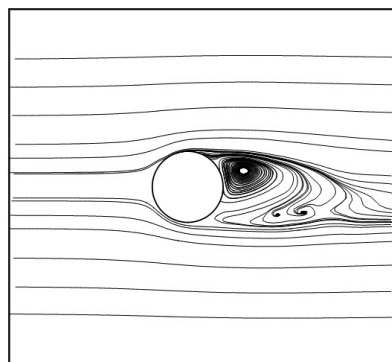
*The first row of the method-III denotes the total elapsed time for the completion of one time step while the second row the elapsed time for solving SPTM in step III.

Table 7. Comparison of CPU times (in second) of the three methods for the simulation of the unsteady flow past a sphere at $Re = 300$ from $tU/d = 0.0$ to $tU/d = 1.0$ when four different time step size are used.

Time-step (CFL No.)	0.005 (0.54)	0.01	0.025	0.05 (5.4)
Method-I	12276.3	7824.6	3517.8	X
Method-II	2935.6	X	X	X
Method-III	6513.9	3431.5	1739.3	904.0



(a) (x, y)-plane



(b) (z, x)-plane

Fig. 13. Streamlines around a sphere at $Re = 300$.

from $tU/d = 0.0$ to $tU/d = 1.0$ for four time-steps. It reaffirms that Method-III is more efficient than the other methods in terms of the CPU time and stability.

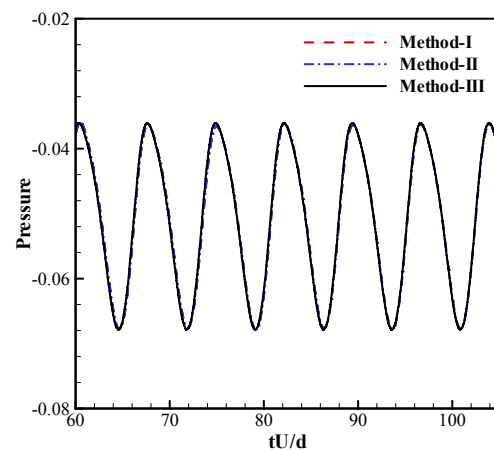


Fig. 14. Temporal evolution of pressure at a point behind a sphere at $Re = 300$.

4. Conclusions

A fractional three-step mixed finite element method (method-III), which is a $P2P1$ finite element version of the $P1P1$ four-step fractional step method [1], was compared with conventional one-step integrated methods in terms of the CPU time, convergence characteristics of an iterative solver, and stability.

Regarding one-step integrated schemes, both the Crank-Nicolson (method-I) and Adams-Bashforth (method-II) methods were employed for comparison. From the unsteady numerical experiments of the proposed methods, the following conclusions were drawn:

1) The iteration number of Bi-CGSTAB for the solution of the momentum equations (the first step) of method-III is very small

compared to those for the solution of the saddle-point type matrices (SPTM) of method-I and method-II. On the other hand, the non-linear iteration number of method-III is close to that of method-I regardless of the time-step used. Therefore, most of the CPU time of method-III is consumed for solving the SPTM in the third step for unsteady flow around a cylinder.

2) Method-III is about 2 times faster than method-I because the SPTM is assembled only once during the entire computation and solved once at each time-step.

3) Method-III is shown to be more stable than the other methods allowing 10 times larger time-step than method-II while having comparable accuracy. Method-II is stable only when the CFL number is less than 1.0.

4) Considering the three conclusions above, it can be said that method-III is more efficient than the other methods in terms of the CPU time and stability.

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Nomenclature

u	: Velocity vector
p	: Pressure
σ	: Stress tensor
μ	: Dynamic viscosity
ρ	: Density
W	: Test functions
q	: Weighting function of the continuity equation
Re	: Reynolds number
τ	: Shear stress
Δt	: Time increment
St	: Strouhal number

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